Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes

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Abstract

Automatic differentiation (AD) has driven recent advances in machine learning, including deep neural networks and Hamiltonian Markov Chain Monte Carlo methods. Partially observed nonlinear stochastic dynamical systems have proved resistant to AD techniques because widely used particle filter algorithms yield an estimated likelihood function that is discontinuous as a function of the model parameters. To resolve this, we create a theoretical framework that embeds two existing AD particle filter methods within a new class of algorithms. This new class permits a bias-variance tradeoff and hence a mean squared error substantially lower than the existing algorithms. This allows us to develop likelihood maximization algorithms suited to the Monte Carlo properties of the AD gradient estimate. Our algorithms require only a differentiable simulator for the latent dynamic system; by contrast, most previous approaches to AD likelihood maximization for particle filters require access to the system's transition probabilities. Numerical results indicate that a hybrid algorithm that uses AD to refine a coarse solution from an iterated filtering algorithm shows substantial improvement on current state-of-the-art methods on a challenging scientific benchmark problem.

1 Introduction

Many scientific models involve highly nonlinear stochastic dynamic systems possessing significant random variation in both the process dynamics and the measurements. Commonly, the latent system is modeled as a Markov process, giving rise to a partially observed Markov process (POMP) model, also known as a hidden Markov models or a state space model. POMP models arise in fields as diverse as optimal control (Singh et al., 2023), epidemiology (He et al., 2010; Stocks et al., 2018), ecology (Knape and de Valpine, 2012) and finance (Kim and Stoffer, 2008; Bretó, 2014). Despite their ubiquity, estimation and inference for this broad class of models remains a challenging problem. We develop full-information plug-and-play inference for POMP models that takes advantage of automatic differentiation (AD) computation while controlling the Monte Carlo bias and variance of the resulting estimator. These desiderata have not previously been combined, and we discuss each component in turn.

Full-information methods are those which make statistically efficient use of data, typically via likelihood-based or Bayesian methods that avoid dependence on summary statistics. The particle filter, also known as sequential Monte Carlo, serves as the foundation for various full-information inference algorithms for POMP models, since it provides an unbiased estimate of the likelihood function (Del Moral, 2004), enabling Bayesian inference via particle Markov chain Monte Carlo (Andrieu et al., 2010) and likelihood-based inference via iterated filtering algorithms IF1 (Ionides et al., 2006) and IF2 (Ionides et al., 2015).

A POMP model inference algorithm is said to be plug-and-play if it does not need to evaluate the transition density of the latent Markov process, enabling an arbitrary model simulator to be plugged into the algorithm, a convenient property that facilitates scientific discovery (He et al., 2010). Some plug-and-play methods depend on the construction of low-dimensional summary statistics (Wood, 2010; Toni et al., 2009), sacrificing statistical information for computational convenience. Here, our mission is to extend current computational capability among methods wich maintain statistical efficiency and the plug-and-play property. Plug-and-play methods are sometimes called likelihood-free (Owen et al., 2015), and it may be counter-intuitive that likelihood-based inference is possible using likelihood-free methods. Basic particle filters, also known as boostrap filters, possess the plug-and-play property. Iterated particle filtering algorithms, such as IF1 and IF2, have shown that plug-and-play likelihood maximization is practical in various scientific investigations (King et al., 2008; Blake et al., 2014; Pons-Salort and Grassly, 2018; Subramanian et al., 2021; Fox et al., 2022; Drake et al., 2023). However, iterated filtering algorithms have slow convergence near the MLE (Doucet et al., 2013)

Recent advances in AD for gradient estimation using particle filters (Naesseth et al., 2018; Jonschkowski et al., 2018; Corenflos et al., 2021; Ścibior and Wood, 2021; Singh et al., 2023) have drawn attention to AD as a tool for inference in POMP models. Widespread adoption of these approaches have been limited by their bias (Naesseth et al., 2018; Jonschkowski et al., 2018), high variance (Poyiadjis et al., 2011; Ścibior and Wood, 2021), computational expense quadratic in the number of particles (Corenflos et al., 2021; Chen and Li, 2024), or by the loss of the plug-and-play property (Poyiadjis et al., 2011; Ścibior and Wood, 2021; Singh et al., 2023; Chen and Li, 2024). We present a new algorithm which (for reasons which will be discussed later) we call a *Differentiated Measurement Off-Parameter* (DMOP) particle filter. DMOP corresponds to the derivative of a smooth variant of the particle filter that we call MOP, though DMOP is conveniently calculated using AD without the need to actually implement the full MOP filter. MOP has the special property that, unlike most particle filters, the gradient of the filtered log-likelihood estimate provides a consistent estimate of the gradient of log-likelihood. Particle filters have previously been adapted to provide a smooth estimate of the likelihood function (Svensson et al., 2018; Malik and Pitt, 2011), but MOP and DMOP differ from these by possessing the plug-and-play property.

DMOP trades bias against variance via a discount factor, $\alpha \in [0, 1]$. We show that DMOP-α posesses a bias-variance tradeoff that permits both low bias and low variance for suitable choices of α strictly in $(0, 1)$, both in theory (Theorem 5) and practice (Figure 1). We show that DMOP-α coincides with the biased gradient estimator of Naesseth et al. (2018) when $\alpha = 0$. In a limit as the number of particles increases, DMOP-α matches the high-variance gradient estimate from (Poyiadjis et al., 2011; Ścibior and Wood, 2021) when $\alpha = 1$ (Theorem 1). At $\alpha = 1$, the variance can become prohibitively high, as discovered by these researchers and others (Arya et al., 2022). However, the variance rapidly reduces as α decreases, and so a choice of α close to one can provide a low-bias, low-variance, plug-and-play gradient estimate with computational scaling linear in the number of particles.

In practice, likelihood optimization via gradient descent with DMOP-α converges quickly within a neighborhood of the maximum but struggles to reach this neighborhood in the presence of local minima and saddle points. By contrast, iterated filtering algorithms provide a relatively fast and stable way to identify this neighborhood. We therefore build a simple but effective hybrid algorithm called *Iterated Filtering with Automatic Differentiation* (IFAD) that warm-starts first-order or second-order gradient methods with a preliminary solution obtained from iterated filtering. We implement our algorithm using JAX (Bradbury et al., 2018), which also enables us to take advantage of graphical processing unit (GPU) hardware. Our methodology is provided in an open-source software package, `pypomp` (Abkemeier et al., 2025).

Promising numerical results indicate that IFAD beats IF2 (and by the numerical results of Ionides et al. (2015), also IF1 and the Liu-West filter (Liu and West, 2001)) on a challenging problem in epidemiology, the Dhaka cholera model of King et al. (2008). Empirically, the benchmark tests demonstrate that IFAD is the most significant methodological advance since Ionides et al. (2015) for plug-and-play statistical inference on POMP models. The choice of models and methods for benchmarking is justified in Section S8.

We also demonstrate empirical advances using DMOP-α for Bayesian inference. In Section 6, we use the DMOP-α gradient estimates within a No-U-Turn Sampler (NUTS) (Homan and Gelman, 2014) in conjunction with an empirical Bayes-type prior estimated with IF2 to reduce the burn-in period of particle MCMC (Andrieu et al., 2010) on the Dhaka cholera model of King et al. (2008) from the $700,000$ iterations in Fasiolo et al. (2016) to just 500. To the best of our knowledge, attaining rapid mixing on this challenging problem has not previously been achieved.

2 Problem Setup

Consider an unobserved Markov process $\{X(t), t \geq t_0\}$, with discrete-time observations $Y_1, ..., Y_N$ realized at values $y_1^*, \ldots, y_N^*$ with times $t_1, \ldots, t_N$. The process is parameterized by an unknown parameter $\theta \in \Theta \subseteq \mathbb{R}^p$, where the state $X(t)$ takes values in the state space $\mathcal{X} \subseteq \mathbb{R}^d$, the observations Y_n take values in $\mathcal{Y}$, and we write $X_n = X(t_n)$. We suppose that the discrete-time latent process model has a density $f_{X_n|X_{n-1}}(x_n \mid x_{n-1}; \theta)$. The existence of this density is necessary to define the likelihood function, but plug-and-play algorithms do not explicitly evaluate it. Instead, they have access to a corresponding simulator, which write as $\texttt{process}_n(x_n \mid x_{n-1}; \theta)$. We set $f_{Y_n|X_n}(y_n \mid x_n; \theta)$ to be the measurement density, which can be directly evaluated. We call $f_{X_{1:n}|Y_{1:n}}(x_{1:n}|y_{1:n}^*; \theta)$ the posterior distribution of states, and $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$ the filtering distribution at time t_n. All joint and conditional densities are defined on a probability space $(\Omega, \Sigma, \mathbb{P})$ which is also assumed to enable construction of independent replicates and all random variables defined in our algorithms. For the algorithmic interpretation of our theory, we identify the random number seed with an element of the sample space $\omega \in \Omega$. The random seed determines the sequence of pseudo-random numbers generated by a computer, just as $\omega \in \Omega$ generates the sequence of random variables.

3 Off-Parameter Particle Filters

The *Measurement Off-Parameter with discount factor* α (MOP-α), particle filter is defined by the pseudocode in Algorithm 1. Note that this algorithm has the plug-and-play property. We suppose that the simulator is a differentiable function of θ for every fixed ω. This condition is equivalent to the *reparameterization trick* (Corenflos et al., 2021); by writing the generated random numbers as parameter-dependent transformation from a reference distribution—e.g., the quantile transform of a uniform random variable—we can differentiate their values, and hence the simulation, with respect to parameters. The discrete resampling step of the bootstrap particle filter has been the primary point of concern for the use of AD to estimate derivatives for particle filters. The discontinuity of this resampling (as a function of θ, for fixed ω) means that the derivative of the expectation is not consistently estimated by the expectation of the derivative. MOP-α is a mathematical construction designed to avoid this discontinuity. We begin with a filter that resamples particles with weights calculated at a baseline parameter value, ϕ, and then corrects these weights to target the filtering distribution at another parameter value, θ, yielding the name *measurement off-parameter*. The name is chosen by analogy with similar *off-policy* algorithms used for reinforcement learning in which a policy that was not implemented is evaluated by reweighting the rewards of the policy that was actually applied. If we fix ω and ϕ, the sample indices do not change with θ, but the re-weighting scheme does, allowing MOP-α to be differentiated with respect to θ. When $\theta = \phi$, we have $w^{P,\theta}_{n,j} = w^{F,\theta}_{n-1,j} = 1$ and MOP-α reduces to the standard particle filter; this is the setting we use in practice, but use the construction allows us to define a derivative. One can think of MOP-α as reconstructing local smooth approximations of the discontinuous likelihood estimates produced by a particle filter for a fixed number of particles J and fixed seed ω. These approximate the likelihood at θ in a neighborhood around ϕ, with the intent that differentiating these smooth log-likelihood reconstructions yields a score estimate.

DMOP-α (Algorithm 2) computes the fixed-seed derivative of the likelihood estimate obtained by MOP-α at $\theta = \phi$. To better understand the implementation DMOP-α we review some background for AD. Standard AD consists of two evaluation passes: a forward pass in which the function is evaluated and a graph built of the evaluation process, followed by a backward pass where the derivative is evaluated by recursive use of the chain rule on the computation graph (Griewank and Walther, 2008). The two evaluation passes in MOP-α at ϕ and θ correspond directly to the two evaluation passes in DMOP-α. Differentiation of MOP-α with respect to θ requires a preliminary evaluation at ϕ and subsequent differentiation with respect to θ for the second pass which we obtain through AD. In practice, however, we only work in the $\theta = \phi$ setting in which case MOP-α corresponds to the standard particle filter and we only need to make one pass. In order to apply AD in this setting without using two passes, we must both instantiate the weighting scheme (even though the weights are always 1) and distinguish which components of the weights correspond to which pass. This can be achieved using the *stop-gradient* capability that appears in Algorithm 2 where its application in the weight correction step indicates that AD should apply the chain rule to the numerator but not the denominator of the correction; note that it is this construction that yields weights which are always 1, but which also have non-zero derivatives.

MOP-α has similarities with Svensson et al. (2018), in which particles from a previous run at ϕ are reweighted to evaluate the likelihood at any sufficiently nearby parameter θ for 0-th order optimization. However, their method is not plug-and-play because they use transition densities to reweight particles, and the variance of their log-likelihood estimate has an exponential dependence in the horizon, N, when $\theta \neq \phi$. Our approach bypasses these issues – the first since the reparameterization trick allows us to consider particle paths dependent on θ, and the second as using AD allows us to only evaluate the likelihood and score estimates when $\theta = \phi$.

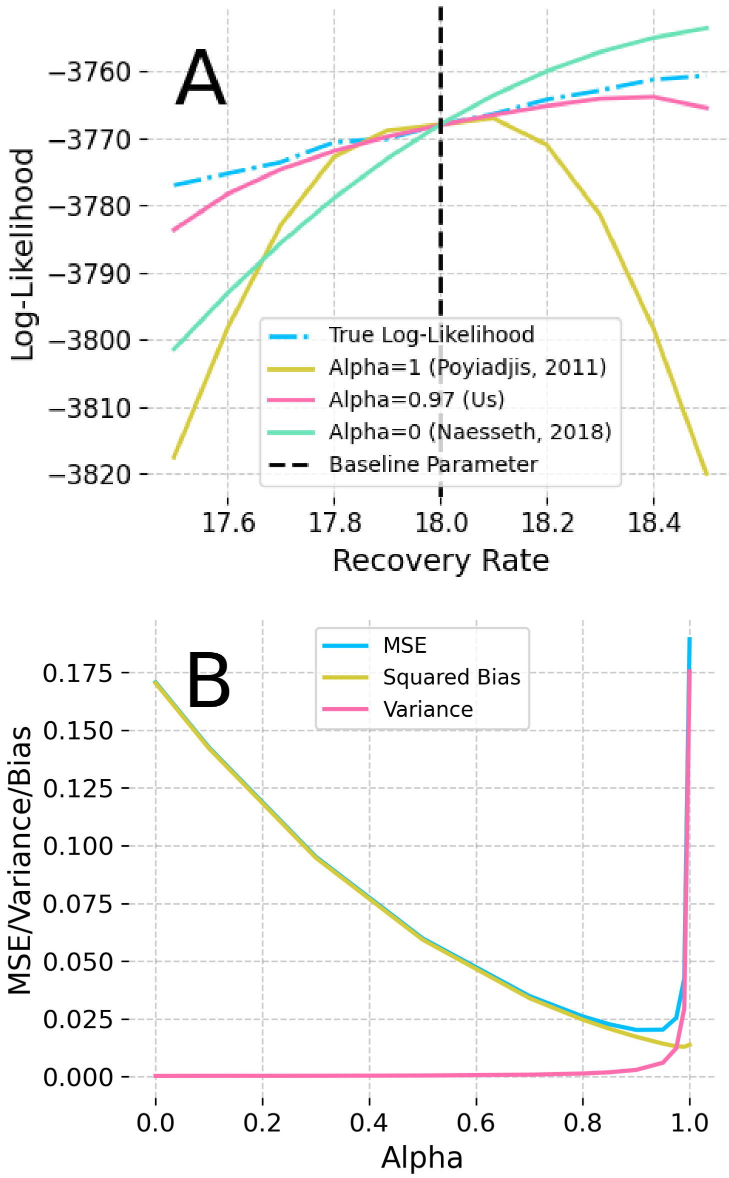

Figure 1: **A:** Non-differentiable likelihood estimates and smooth off-parameter approximations that hold locally around the baseline parameter ϕ. **B:** Illustration of the bias-variance tradeoff induced by the discounting hyperparameter α. We display the MSE of score estimates for the trend in transmission at the MLE for the Dhaka cholera model of King et al. (2008). Our novel contribution of $\alpha \in (0, 1)$ significantly improves the quality of the likelihood approximation and the score estimates.

Algorithm 1 MOP-α

Input: Number of particles J, timesteps N, measurement model $f_{Y_n|X_n}(y_n^*|x_n,\theta)$, simulator $\texttt{process}_n(\cdot|x_n;\theta)$, evaluation parameter θ, baseline parameter ϕ, seed ω.
First pass: Set $\theta=\phi$ and fix ω, yielding $X_{n,j}^{P,\phi}$, $X_{n,j}^{F,\phi}$, $g_{n,j}^{\phi}$.
Second pass: Fix ω, and filter at $\theta\neq\phi$:
Initialize particles $X_{0,j}^{F,\theta}\sim f_{X_0}(\cdot\,;\theta)$, weights $w_{0,j}^{F,\theta}=1$.
For $n=1,...,N$:
  Accumulate discounted weights, $w_{n,j}^{P,\theta}=\big(w_{n-1,j}^{F,\theta}\big)^{\alpha}$.
  Simulate process model, $X_{n,j}^{P,\theta}\sim\texttt{process}_n\big(\cdot\,|X_{n-1,j}^{F,\theta};\theta\big)$.
  Measurement density, $g_{n,j}^{\theta}=f_{Y_n|X_n}(y_n^*|X_{n,j}^{P,\theta};\theta)$.
  Compute $L_n^{B,\theta,\alpha}=\sum_{j=1}^J g_{n,j}^{\theta}\,w_{n,j}^{P,\theta}\,/\sum_{j=1}^J w_{n,j}^{P,\theta}$.
  Conditional likelihood under ϕ, $L_n^{\phi}=\frac{1}{J}\sum_{m=1}^J g_{n,m}^{\phi}$.
  Select resampling indices $k_{1:J}$ with $\mathbb{P}\big(k_j=m\big)\propto g_{n,m}^{\phi}$.
  Obtain resampled particles $X_{n,j}^{F,\theta}=X_{n,k_j}^{P,\theta}$.
  Calculate corrected weights $w_{n,j}^{F,\theta}=w_{n,k_j}^{P,\theta}\,g_{n,k_j}^{\theta}\,/\,g_{n,k_j}^{\phi}$.
  Compute $L_n^{A,\theta,\alpha}=L_n^{\phi}\cdot\sum_{j=1}^J w_{n,j}^{F,\theta}\,/\sum_{j=1}^J w_{n,j}^{P,\theta}$.
Return: likelihood estimate $\hat{\mathcal{L}}^A(\theta)=\prod_{n=1}^N L_n^{A,\theta,\alpha}$ or $\hat{\mathcal{L}}^B(\theta)=\prod_{n=1}^N L_n^{B,\theta,\alpha}$, and filtering distributions $\{(X_{n,j}^{F,\theta},w_{n,j}^{F,\theta})\}_{n,j=1}^{N,J}$.

Algorithm 2 DMOP-α

Input: Number of particles J, timesteps N, measurement model $f_{Y_n|X_n}(y_n^*|x_n,\theta)$, simulator $\texttt{process}_n(\cdot|x_n;\theta)$, parameter θ.
Forward pass: AD constructs a computation graph for the likelihood estimate, $\hat{\mathcal{L}}^A(\theta)=\prod_{n=1}^N L_n^{A,\theta,\alpha}$ or $\hat{\mathcal{L}}^B(\theta)=\prod_{n=1}^N L_n^{B,\theta,\alpha}$.
Backward pass: AD evaluates the gradient estimate, $\hat{\mathcal{D}}^A(\theta)$ or $\hat{\mathcal{D}}^B(\theta)$.
Initialize particles $X_{0,j}^{F,\theta}\sim f_{X_0}(\cdot\,;\theta)$, weights $w_{0,j}^{F,\theta}=1$.
For $n=1,...,N$:
  Accumulate discounted weights, $w_{n,j}^{P,\theta}=\big(w_{n-1,j}^{F,\theta}\big)^{\alpha}$.
  Simulate process model, $X_{n,j}^{P,\theta}\sim\texttt{process}_n\big(\cdot\,|X_{n-1,j}^{F,\theta};\theta\big)$.
  Measurement density, $g_{n,j}^{\theta}=f_{Y_n|X_n}(y_n^*|X_{n,j}^{P,\theta};\theta)$.
  Compute $L_n^{B,\theta,\alpha}=\sum_{j=1}^J g_{n,j}^{\theta}\,w_{n,j}^{P,\theta}\,/\sum_{j=1}^J w_{n,j}^{P,\theta}$.
  Select resampling indices $k_{1:J}$ with $\mathbb{P}\big(k_j=m\big)\propto g_{n,m}^{\theta}$.
  Obtain resampled particles $X_{n,j}^{F,\theta}=X_{n,k_j}^{P,\theta}$.
  Calculate corrected weights $w_{n,j}^{F,\theta}=w_{n,k_j}^{P,\theta}\,g_{n,k_j}^{\theta}\,/\,\texttt{stop\_gradient}(g_{n,k_j}^{\theta})$.
  Compute $L_n^{A,\theta,\alpha}=L_n^{\phi}\cdot\sum_{j=1}^J w_{n,j}^{F,\theta}\,/\sum_{j=1}^J w_{n,j}^{P,\theta}$.
Return: likelihood estimate $\hat{\mathcal{L}}^A(\theta)=\prod_{n=1}^N L_n^{A,\theta,\alpha}$ or $\hat{\mathcal{L}}^B(\theta)=\prod_{n=1}^N L_n^{B,\theta,\alpha}$, and gradient estimate $\hat{\mathcal{D}}^A(\theta)$ or $\hat{\mathcal{D}}^B(\theta)$.

3.1 Some algorithmic details

The resampling indices, which we write as $k_j \sim \text{Categorical}(g^{\phi}_{n,1}, ..., g^{\phi}_{n,J})$, are a function of ϕ for any value of θ. For the categorical distribution, we use systematic resampling (Arulampalam et al., 2002; King et al., 2016) which usually outperforms multinomial resampling.

In Algorithm 1, the conditional likelihoods are reweighted by a correction factor $w^{P,\theta}_{n,j}$ that accumulates over filtering steps to account for the resampling under ϕ. Writing $g^{\theta}_{n,j} = f_{Y_n|X_n}(y^*_n|X^{P,\theta}_{n,j}; \theta)$ for the measurement density and $L^{\phi}_n = \frac{1}{J}\sum_{m=1}^{J} g^{\phi}_{n,m}$ for the conditional likelihood estimate under ϕ, we obtain two estimates of the conditional likelihood under θ,

$$L^{B,\theta,\alpha}_n = \frac{\sum_{j=1}^{J} g^{\theta}_{n,j}\, w^{P,\theta}_{n,j}}{\sum_{j=1}^{J} w^{P,\theta}_{n,j}}, \qquad L^{A,\theta,\alpha}_n = L^{\phi}_n \cdot \frac{\sum_{j=1}^{J} w^{F,\theta}_{n,j}}{\sum_{j=1}^{J} w^{P,\theta}_{n,j}}, \tag{1}$$

where we recursively correct the weights for each particle for the cumulative error incurred by the off-parameter resampling:

$$w^{P,\theta}_{n,j} = (w^{F,\theta}_{n-1,j})^{\alpha}, \quad w^{F,\theta}_{n,j} = w^{P,\theta}_{n,k_j}\, g^{\theta}_{n,k_j} \,/ g^{\phi}_{n,k_j}, \quad w^{F,\theta}_{0,j} = 1. \tag{2}$$

The before-resampling estimate $L^{B,\theta,\alpha}_n$ is preferable in practice, as it has slightly lower variance than the after-resampling estimate $L^{A,\theta,\alpha}_n$, but the latter is useful in deriving properties of the MOP-α gradient estimate such as that in Theorem 1.

3.2 Discounted Weights and a Bias-Variance Tradeoff

Heuristically, α controls the exponentially decaying memory of MOP and DMOP. Importantly, α is irrelevant for MOP at $\theta = \phi$ when the weights are always 1 and therefore also irrelevant for the function evaluation in the first pass of DMOP, but crucially not in the second when the gradients are evaluated. If $\alpha = 1$, the gradient of the weights, $w^{P,\theta}_{n,j}$ and $w^{F,\theta}_{n,j}$, accumulates as n increases. This leads to numerical instability and high variance unless N is small. For $\alpha < 1$, the weights from previous timesteps are discounted as in Equation (2), allowing for lower variance at the expense of bias in the gradient estimate.

When $\alpha = 1$, MOP-1 maintains complete memory of each particle's ancestral trajectory. Theorem 1 shows that DMOP-1 recovers the consistent but high-variance score estimate of Poyiadjis et al. (2011). When $\alpha = 0$, MOP-0 considers only single-step transition dynamics, and DMOP-0 recovers the low-variance but asymptotically biased score estimate of Naesseth et al. (2018). Ścibior and Wood (2021) showed that this matches the gradient of a vanilla particle filter when resampling terms are dropped.

Theorem 1 (DMOP-0 and DMOP-1 Functional Forms)**.** *Writing $\nabla_\theta \hat{\ell}^{\alpha}(\theta)$ for the gradient estimate yielded by DMOP-α and using the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^{N} L^{A,\theta,\alpha}_n$, when $\alpha = 0$,*

$$\nabla_\theta \hat{\ell}^{0}(\theta) = \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J} \nabla_\theta \log f_{Y_n|X_n}(y^*_n|x^{F,\theta}_{n,j}; \theta),$$

yielding Naesseth et al. (2018)'s estimator with the bootstrap filter. When $\alpha = 1$,

$$\nabla_\theta \hat{\ell}^{1}(\theta) = \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y^*_{1:N}|x^{A,F,\theta}_{1:n,j}\right),$$

yielding the estimator of Poyiadjis et al. (2011) for the bootstrap filter.

We defer the proof to Appendix **??**. The argument relies on a useful decomposition of the after-resampling estimate $L_n^{A,\theta,\alpha}$, repeated applications of the log-derivative identity that $\nabla_x \log(f(x)) = (\nabla_x f(x))/f(x)$, and noting $\theta = \phi$ implies that $w_{n,j}^{P,\theta}$ evaluates to 1.

The result in Theorem 1 provides mathematical intuition for why Ścibior and Wood (2021)'s use of the stop-gradient procedure in the particle filter recovers the estimate of Poyiadjis et al. (2011). When $\theta = \phi$, the correction $g_{n,k_j}^{\theta}/g_{n,k_j}^{\phi}$ is equivalent to $g_{n,k_j}^{\theta}/\text{stop\_gradient}(g_{n,k_j}^{\theta})$, and is implemented in this way in practice. As such, the use of the stop-gradient in Ścibior and Wood (2021) can be seen as the MOP-α weight correction for $\alpha = 1$, and $\theta = \phi$.

4 From Gradients to Maximum Likelihood Estimation

To design an effective procedure for likelihood maximization using DMOP-α, we carry out iterated filtering then automatic differentiation (IFAD). A basic IFAD approach is presented in Algorithm **??**. This requires a Hessian estimate $H(\theta)$ with minimum eigenvalue no smaller than c, but we allow for the possibility of a trivial estiamte, $H = I_p$, in which case the AD search becomes stochastic gradient descent (SGD). Algorithm **??** is convenient for theoretical analysis, but two further modifications are helpful in practice. Our best numerical results were obtained by putting the gradient estimate into the Adam optimizer (Kingma and Ba, 2015) together with learning rate decay. Numerical results are provided in Section 5.

Algorithm 3 IFAD

Input: Number of particles J, timesteps N, IF2 and MOP-α cooling schedules η_m, MOP-α discounting parameter α, θ_0, $m = 0$.
Run initial IF2 search under cooling schedule η_m to obtain $\{\Theta_j, j = 1, ..., J\}$, set $\theta_m := \frac{1}{J}\sum_{j=1}^{J}\Theta_j$.
While procedure not converged:
  Run Algorithm 1 to obtain $\hat{\ell}(\theta_m)$.
  Set $g(\theta_m) := \nabla_\theta(-\hat{\ell}(\theta_m))$, and consider any $H(\theta_m)$ s.t. $\lambda_{\min}(H(\theta_m)) \geq c$.
  Update $\theta_{m+1} := \theta_m - \eta_m(H(\theta_m))^{-1} g(\theta_m)$, $m := m + 1$.
Return $\hat{\theta} := \theta_m$.

The convergence of IF2 (and so the first stage of IFAD) to a neighborhood of the MLE happens rapidly in practice. However, even when applying much larger amounts of Monte Carlo effort, IF2 fails to get within seven ADJUST FOR BEST IF2 AND NEW UNDERSTANDING OF MLE log-likelihood units of the maximum on this problem (Figure 4). For this example, IF2, without assistance from AD, struggles to reach the inferentially critical region within one log-likelihood unit of the maximum, even when additional computationally intensive techniques are used, such as repeated pruning and profiling,

The AD stage confers a linear convergence rate for IFAD, up to a small region near the MLE shrinking in J, under the usual smoothness and strong convexity assumptions in convex optimization theory, as we show below in Theorem 2.

Theorem 2 (Linear Convergence of IFAD)**.** *Assume strongly convex and smooth $-\ell$, $\gamma I \preceq \nabla_\theta^2(-\ell) \preceq \Gamma I$, and assumptions (A1–A5) in Section 7. Choose the learning rate η so $\eta \leq \frac{c(1-\beta)}{2\Gamma}$. Consider the second stage of IFAD (Algorithm 3), stopping if $\|\nabla_\theta \hat{\ell}^\alpha(\theta_m)\| \leq (1+\sigma)\epsilon$, where $\sigma \geq \frac{4\Gamma}{(1-\beta)}$ for some $\beta \in (0, 1)$. Choose the learning rate η so $\eta \leq \frac{c(1-\beta)}{2\Gamma}$. Then, for α sufficiently* *close to 1 and J sufficiently large (depending on ϵ and δ), we have that $\|\nabla_\theta \hat{\ell}^\alpha(\theta_m) - \nabla_\theta \ell^\alpha(\theta_m)\| \leq \epsilon$ and $H(\theta_m) \succ cI_p \succ 0$ for all m, and that*

$$\ell(\theta^*) - \ell(\theta_{m+1}) \leq \Big(1 - \eta\beta\,\frac{8\gamma}{9c}\Big)\big(\ell(\theta^*) - \ell(\theta_m)\big) \text{ with probability at least } 1-\delta.$$

The proof of Theorem 2 uses results from randomized numerical linear algebra, building on Theorem 6 of Roosta-Khorasani and Mahoney (2016). Details are deferred to Appendix **??**. This result shows that particle estimates for the gradient can enjoy the same linear convergence rate on well-conditioned problems as gradient descent with access to the score.

We therefore see that the second stage of IFAD converges linearly to the MLE if (1) the log-likelihood surface is well-conditioned in a neighborhood of the MLE and (2) the first stage of IFAD successfully reaches a (high-probability) basin of attraction of the MLE. Within a neighborhood of the MLE that grows as more data become available (Ning et al., 2021), local quadratic approximations such as local asymptotic normality apply under general conditions (Le Cam and Yang, 2000). This suggests that the linear convergence may occur often in practice.

5 Application to a Cholera Transmission Model

The cholera transmission model that King et al. (2008) developed for Dhaka, Bangladesh, has been used to benchmark the performance of various POMP inference methods (Ionides et al., 2015; Fasiolo et al., 2016; Wycoff et al., 2026), and we employ it here for the same purpose. This model categorizes individuals in a population as susceptible, $S(t)$, infected, $I(t)$, and recovered, $R(t)$ and so is called an SIRS compartmental model. In this case, the compartment $R(t)$ is further subdivided into three recovered compartments $R^1(t)$, $R^2(t)$, $R^3(t)$ denoting varying degrees of cholera immunity. We write $P(t)$ for the total population, and M_n for the cholera deaths in each month. As in King et al. (2008), the transition dynamics follow a stochastic differential equation (SDE):

$$\begin{aligned}
dS &= \big(k\epsilon R^k + \delta(P-S) - \lambda(t)S\big)\,dt + dP - (\sigma SI/P)\,dB,\\
dI &= \big(\lambda(t)S - (m+\delta+\gamma)I\big)\,dt + (\sigma SI/P)\,dB,\\
dR^1 &= \big(\gamma I - (k\epsilon+\delta)R^1\big)\,dt, \qquad \dots\\
dR^k &= \big(k\epsilon R^{k-1} - (k\epsilon+\delta)R^k\big)\,dt,
\end{aligned}$$

with Brownian motion $B(t)$, cholera death rate m, recovery rate γ, mean immunity duration $1/\epsilon$, standard deviation of the force of infection σ, and population death rate $\delta = 0.02$. The force of infection, λ_t, is modeled by splines $(s_j)_{j=1}^6$

$$\lambda_t = \exp\left(\beta_{\text{trend}}(t-t_0) + \sum_{j=1}^{6}\beta_j s_j(t)\right)\frac{I}{P} + \exp\left(\sum_{j=1}^{6}\omega_j s_j(t)\right),$$

where the coefficients $(\beta_j)_{j=1}^6$ model seasonality in the force of infection, β_{trend} models the trend in the force of infection, and the ω_j represent seasonality of a non-human environmental reservoir of disease. The measurement model for observed monthly cholera deaths is $Y_n \sim \mathcal{N}(M_n, \tau^2 M_n^2)$, where $M_n = \gamma \int_{t_{n-1}}^{t_n} I(s)\,ds$ is the modeled cholera deaths in that month.

We numerically solve this SDE using Euler's method with a time step of 1/20 months, matching (King et al., 2008). This example is a partially observed hypo-elliptical SDE, and methodology specific to this model class has been previously studied (Ditlevsen and Samson, 2019). Our method applies to a class of latent Markov processes larger than Gaussian SDEs, permitting the use of Lévy noise (Bhadra et al., 2011) or other discontinuities.

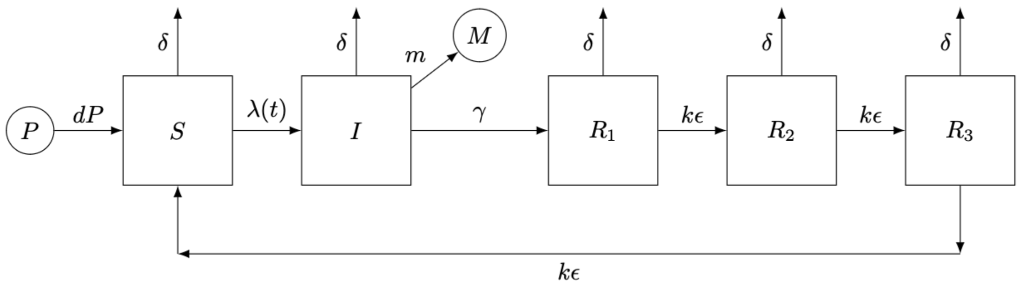

Figure 2: A compartment flow diagram for the King et al. (2008) Dhaka cholera model.

We tested IFAD against IF2 on a global search problem for the Dhaka cholera model. We re-implemented IF2 to do so, but we also compare our results with the results of Ionides et al. (2015) (labeled "IF2 2015"). Our re-implementation reaches a better log-likelihood than that of Ionides et al. (2015) due to a better choice of algorithmic parameters. For each method, we performed 100 searches, initialized with 100 starting parameter vectors drawn uniformly from the same wide bounding box used in Ionides et al. (2015). We summarize our findings below.

	Best Log-Likelihood	Rank
IFAD-0.97	-3744.17	1
IFAD-1	-3745.12	2
IF2	-3747.82	3
IFAD-0	-3748.69	4

Table 1: Maximum log-likelihood found by IF2 and IFAD. IFAD-0.97 performs the best among all methods.

Previously, an MLE at a log-likelihood of -3748.6 was reported by King et al. (2016). This MLE was obtained with much computational effort, using many global and local IF2 searches and with the assistance of likelihood profiling, 8 years after the model was first proposed by King et al. (2008). Meanwhile, Ionides et al. (2015) only achieve a maximum log-likelihood of -3768.6, and King et al. (2008) only -3793.4. The results of Table 1, Figure 3 and Figure 4) indicate that the actual MLE is around XXXXX. IFAD has effectively solved a long-standing benchmark problem.

While IF2 quickly approaches a neighborhood of the MLE within only 40 iterations, IF2 alone ultimately fails to achieve the last few log-likelihood units, as no IF2 search comes within 7 XXXX log-likelihood units of the MLE (as seen in Figure 3). Conversely, we encountered many failed searches with gradient descent alone. This is a difficult, nonconvex, and noisy problem with 18 parameters, and gradient descent without a warm-start gets stuck in local minima and saddle points.

IFAD, in comparison, approaches the MLE quickly due to the IF2 warm-start (as seen in Figure 5) and also succeeds at refining the coarse solution found by the warm-start with DMOP gradient steps to find the MLE (as seen in Figures 3 and 4). IFAD therefore successfully combines the best qualities of IF2 and DMOP, outperforming either of them alone. Monte Carlo replication is appropriate for IFAD, or any Monte Carlo algorithm used to solve a challenging numerical problem, but Figure 4 shows that IFAD can find higher likelihood values, more quickly and more reliably than the previous state-of-the-art. In particular, Figure 4 shows that IFAD with $\alpha = 0.97$ can maximize the challenging likelihood of King et al. (2008) using a modest number of iterations and Monte Carlo replications, whereas previously it was necessary to carry out an extensive customized search or to live with an incompletely maximized likelihood. Since replicated searches are innevitably needed for Monte Carlo maximization of this challenging likelihood maximization task, it is necessary to focus on the right tail of the distributions in Figure 4 to compare the capabilities of the methods.

I GUESS AT THIS POINT WE JUST SHOW THE FULL RESULTS WITH ADAM AND COOLING. For the results presented here, we did not employ the many common heuristics used in the machine learning and optimization literature in the gradient descent stage. We used constant learning rates of $0.01, 0.05$, and 0.2 for IFAD-1, 0, and 0.97 respectively, and a constant cooling rate of 0.95 for our IF2 implementation. While techniques such as annealing learning rates, gradient normalization, and momentum could further improve the performance of IFAD in other simulations we performed, we chose to report the results of the simplest implementation to serve as a baseline for the method's performance. Our experimentation with second-order optimization methods, using a Hessian estimate obtained by automatic differentiation of DMOP-α, was computationally intensive and did not yield improved results, likely due to Monte Carlo error in the Hessian estimate and non-convexity of the log-likelihood.

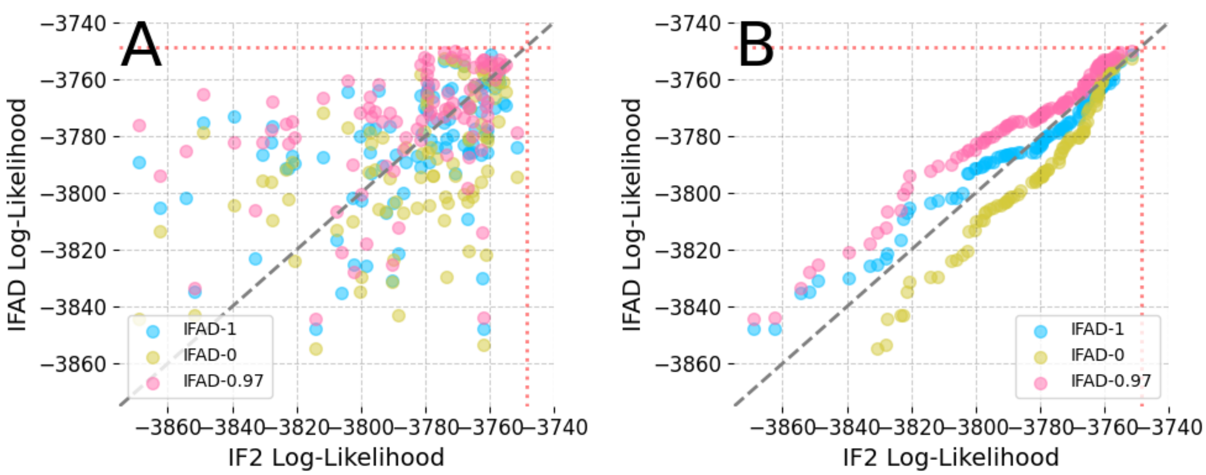

Figure 3: Scatterplots depicting the performance of IFAD against that of IF2. **A:** Paired searches from the same starting point. Tuning α allows IFAD-0.97 to strictly improve on IF2, Poyiadjis et al. (2011), and Naesseth et al. (2018), on almost every initialization. **B:** Q-Q plot of ranked IFAD searches against ranked IF2 searches. IFAD performs best on average, and approaches the MLE while no IF2 search successfully gets within 7 log-likelihood units of it. The dotted red line shows the true maximized log-likelihood.

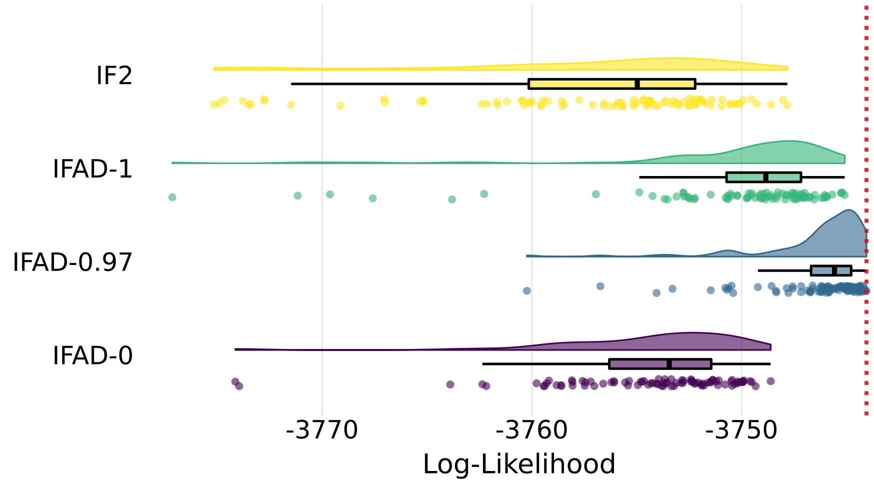

Figure 4: Raincloud plot of all searches. IFAD-0.97 outperforms all other methods. The dotted red line shows the maximized log-likelihood.

6 Application to Bayesian Inference

Particle MCMC (Andrieu et al., 2010) is arguably the most popular method for full-information plug-and-play Bayesian inference for POMP models. However, particle Metropolis-Hastings often mixes slowly, requiring long burn-in periods. For instance, Fasiolo et al. (2016) reported needing 700000 burn-in iterations for effective sampling in the King et al. (2008) model, even after coarsening each Euler timestep to reduce the computational burden. To alleviate this, we employ a No U-Turn Sampler (NUTS) (Homan and Gelman, 2014) powered by DMOP-α, with a nonparametric empirical prior constructed from density estimation of the parameter swarm from the IF2 warm-start. Using NUTS with MOP-α and this prior significantly lowers the mixing time of the Markov chain to as little as 500 iterations (Figure 6). NUTS has been implemented with previous differentiable particle filters (Rosato et al., 2022) but not in our plug-and-play context with control over the bias/variance tradeoff.

Without the IF2 prior, NUTS with the MOP-α estimate alone tended to diverge. In contrast, particle Metropolis-Hastings, with or without said prior, fails to effectively explore the parameter space. We display these results in Figures **??** and **??** in Appendix **??**. This speedup in the mixing time of particle MCMC by over three orders of magnitude provides hope that particle Bayesian inference might be increasingly employed for the complex scientific models commonly encountered in areas like epidemiology.

7 Theoretical Analysis of MOP-α

Here, we will show that MOP-α targets the filtering distribution and likelihood under θ, is consistent when $\alpha = 1$, and characterize rates for its bias and variance under different choices of α. To do so, we require the following assumptions:

(A1) **Continuity of the Likelihood.** $\ell(\theta)$ has more than two continuous derivatives in a neighborhood $\{\theta : \ell(\theta) > \lambda_1\}$ for some $\lambda_1 < \sup_\varphi \ell(\varphi)$.

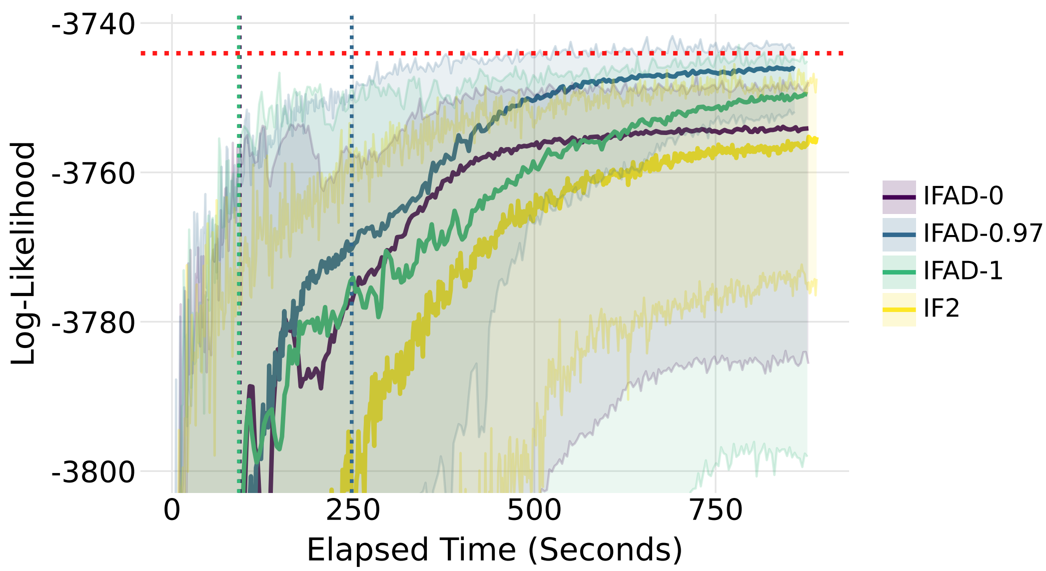

Figure 5: Optimization progress of IFAD and IF2. The shaded regions cover the area between the 100th and 10th percentiles of log-likelihood estimates at each iteration. We use a dotted red line to display the MLE.

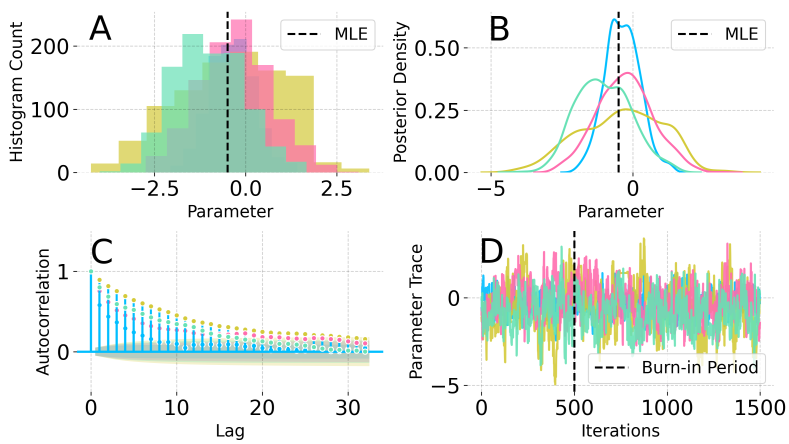

Figure 6: Convergence diagnostics for NUTS with 4 chains, where we plot 1000 samples from the posterior of β_{trend} within the Dhaka cholera model of King et al. (2008). **A:** Histogram of posterior samples. **B:** density estimate of the posterior. **C:** Autocorrelation plot for MCMC samples. **D:** Trace plot for MCMC samples, pre and post burn-in. NUTS mixes quickly when regularized by an empirical prior with a standard deviation of 6.3% of the MLE, and the posterior estimates from each chain share roughly the same posterior mode.

(A2) **Bounded Process Model.** There exist $\underline{M}, \bar{M}$ such that $0 < \underline{M} \leq f_{X_n|X_{n-1}}(x_n|x_{n-1};\theta) \leq \bar{M} < \infty$.

(A3) **Bounded Measurement Model.** There exist $\underline{G}, \bar{G}$ such that $0 < \underline{G} \leq f_{Y_n|X_n}(y_n^* \mid x_n;\theta) \leq$

$\bar{G} < \infty$, and $G'(\theta)$ with $\|\nabla_\theta \log f_{Y_n|X_n}(y_n^* \mid x_n;\theta)\|_\infty \le G'(\theta) < \infty$.

(A4) **Bounded Gradient Estimates.** There are functions $G(\theta), H(\theta) : \Theta \to [0,\infty)$ uniformly bounded by $G^*, H^* < \infty$, so the MOP-α gradient and Hessian estimates at $\theta = \phi$ are almost surely bounded by $G(\theta)$ and $H(\theta)$ for all α.

(A5) **Differentiable Density Ratios and Simulator.** The measurement density, $f_{Y_n|X_n}(y_n^*|x_n;\theta)$, and simulator have more than two continuous derivatives in θ.

In particular, assumptions (A2) and (A3) imply that the POMP model satisfies a strong mixing condition, and as such are fairly strong but commonplace in the literature (Karjalainen et al., 2023; Del Moral, 2004). The fact that the likelihood estimate yielded by MOP-α has more than two continuous derivatives in θ follows from the construction of the likelihood estimate in equations 1 and 2, as well as Assumptions (A3) and (A5).

7.1 MOP-α Targets the Filtering Distribution

We show here that MOP-α targets the filtering distribution under θ and is strongly consistent for the likelihood under θ when $\alpha = 1$ or $\theta = \phi$. Employing $\alpha < 1$ lead to inconsistency when θ deviates from ϕ, but this bias disappears as θ approaches ϕ.

While the result is presented here as specific to MOP-α, we prove a more general result in Appendix **??**. This is a strong law of large numbers for triangular arrays of particles resampled arbitrarily, with resampling probabilities not proportional to the targeted distribution of interest, but reweighted to correct for the cumulative discrepancy between the resampling and the target distribution.

Theorem 3 (MOP-α Targets the Filtering Distribution and Likelihood)**.** *When $\alpha = 1$ or $\theta = \phi$, MOP-α targets the filtering distribution under θ and is strongly consistent for the likelihood under θ. That is, for $\pi_n(\theta) = f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*;\theta)$ and any measurable and bounded functional h and for $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$ or $\prod_{n=1}^N L_n^{B,\theta,\alpha}$, it holds that*

$$\frac{\sum_{j=1}^J h(x_{n,j}^{F,\theta})\, w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{F,\theta}} \overset{a.s.}{\to} E_{\pi_n(\theta)}\big[h(X_n)\big], \qquad \hat{\mathcal{L}}(\theta) \overset{a.s.}{\to} \mathcal{L}(\theta).$$

Proof. We provide a proof sketch here, deferring the details to Appendix **??**. When $\theta = \phi$, the ratio $g_{n,j}^\theta / g_{n,j}^\phi = 1$ regardless of α, reducing to the vanilla particle filter. When $\alpha = 1$ and $\theta \neq \phi$, suppose inductively that $\{(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta})\}_{j=1}^J$ targets $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*;\theta)$. It can then be shown that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}_{j=1}^J$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*;\theta)$, that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta}\, g_{n,j}^\theta)\}_{j=1}^J$ targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*;\theta)$, and that weighting the particles by $(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}) = (X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta}\, g_{n,k_j}^\theta / g_{n,k_j}^\phi)$, resampling the k_j with probabilities proportional to $g_{n,j}^\phi$, also targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*;\theta)$. Estimating the likelihood with the before-resampling estimator $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{B,\theta,\alpha}$, we see that the strong consistency is a direct consequence of our result that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*;\theta)$. $\square$

7.2 DMOP-1 Score Estimate Is Consistent

We take advantage of the equivalence between DMOP-α and the derivative of MOP-α to investigate the former by studying the latter. Although we have shown the DMOP-1 gradient estimate yields the estimate of Poyiadjis et al. (2011) and Ścibior and Wood (2021) when applied to the bootstrap filter, those authors estimate the Fisher score by $\frac{1}{J}\sum_{j=1}^{J}\nabla_\theta \log f_{X_{0:N},Y_{1:N}}(x_{0:n,j}^{A,F,\theta}, y_{1:N}^{*};\theta)$ and not $\frac{1}{J}\sum_{j=1}^{J}\nabla_\theta \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^{*}|x_{1:n,j}^{A,F,\theta};\theta)$. It is therefore not immediately apparent that these two converge to the same thing. As such we directly show the consistency of the MOP-1 gradient estimate below.

Theorem 4 (Consistency of MOP-1 Gradient Estimate)**.** *The gradient estimate of MOP-α when $\alpha = 1$, $\theta = \phi$ is strongly consistent for the score: $\nabla_\theta \hat{\ell}_J^1(\theta) \overset{a.s.}{\to} \nabla_\theta \ell(\theta)$ as $J \to \infty$.*

We present a proof outline here, postponing the full argument to Appendix **??**. Assumption (A4) lets us show uniform equicontinuity of the sequence for almost every $\omega \in \Omega$, allowing us to apply Arzela-Ascoli and Theorem 3 to establish uniform convergence for the derivatives as a sequence $(\nabla_\theta \hat{\mathcal{L}}_J^1(\theta)(\omega))_{J\in\mathbb{N}}$. This is a sufficient condition for swapping the limit and derivative to show that for almost every $\omega \in \Omega$, $\lim_{J\to\infty} \nabla_\theta \hat{\mathcal{L}}_J^1(\theta)(\omega) = \nabla_\theta \lim_{J\to\infty} \hat{\mathcal{L}}_J^1(\theta)(\omega) = \nabla_\theta \mathcal{L}(\theta)$.

We now provide a result showing that DMOP-α has a desirable bias-variance tradeoff when $0 < \alpha < 1$. This is achieved because it combines favorable properties from the low-variance but asymptotically biased estimate of Naesseth et al. (2018) and the high-variance but consistent estimate of Poyiadjis et al. (2011). As the bias itself is difficult to analyze, we instead analyze the MSE and variance. The below result applies for any $\alpha \in [0, 1)$, but not $\alpha = 1$, as the gradient estimate when $\alpha = 1$ has no forgetting properties.

Theorem 5 (MSE and Variance of DMOP-α)**.** *When $\alpha \in (0, 1)$ and $\theta = \phi$, define $\psi_k(\alpha) = (\alpha^k + \alpha^{k+1} - \alpha)/(1 - \alpha)$. There exists an $\epsilon > 0$ depending on $\bar{M}, \underline{M}, \bar{G}, \underline{G}$ as in Karjalainen et al. (2023) such that the MSE and variance of DMOP-α are:*

$$\mathbb{E}\left\|\nabla_\theta \ell(\theta) - \nabla_\theta \hat{\ell}^\alpha(\theta)\right\|_2^2 \;\lesssim\; \min_{k\le N} Np\, G'(\theta)^2 \left(k^2 J^{-1} + (1-\epsilon)^{\lfloor k/(c\log(J))\rfloor} + k + \psi_k(\alpha)\right), \quad (3)$$

$$\mathrm{Var}\left(\nabla_\theta \hat{\ell}^\alpha(\theta)\right) \;\lesssim\; \min_{k\le N} Np\, G'(\theta)^2 \left(\frac{k^2}{(1-\alpha)^2 J} + \frac{\alpha^k}{1-\alpha} N\right). \quad (4)$$

We provide an outline, deferring the full proof to Appendix **??**. The variance bound can be reduced to the approximation error between the score estimate from DMOP-α and a variant called DMOP-(α, k) where the discounted weights are truncated at lag k. The discount factor, α, ensures strong mixing. The covariance of the derivative of MOP-(α, k) are controlled with Davydov's inequality and a standard L^p error bound for the particle filter. The MSE bound considers the error between the target derivative and DMOP-$(1, k)$, and the error between DMOP-$(1, k)$ and DMOP-α. The latter can be shown to be $\tilde{O}\big(Np\, G'(\theta)^2(k + \psi_k(\alpha))\big)$, and the former can be controlled with a result on the the forgetting of the particle filter from Karjalainen et al. (2023) and the very same L^p error bound mentioned above.

Theorem 5 provides theoretical understanding of the empirically favorable bias-variance tradeoff enjoyed by DMOP-α in Figure 1. The first term in the MSE bound in Equation 3 corresponds to particle approximation error, the second mixing error, and the third and fourth the error between the DMOP-$(1, k)$ and DMOP-α score estimates. As α goes to 1, choosing k appropriately, the first, third and fourth term (corresponding to the variance) increases, while the second term (corresponding to the bias) decreases. Likewise, the first term in the variance bound in Equation 4 corresponds to particle approximation error, while the second corresponds to mixing error. The variance increases as α goes to 1.

We show in Appendix **??** that the variance of DMOP-0 is $\tilde{O}\big(Np\,G'(\theta)^2/J\big)$. Previously, Poyiadjis et al. (2011) established that the variance of DMOP-1 is $\tilde{O}(N^4/J)$, ignoring factors of p and G'. Combining these results with Theorem 5 shows that DMOP-α interpolates between the variances of DMOP-0 and DMOP-1, as $N \leq Nk^2 \leq N^4$, with a phase transition as soon as $\alpha < 1$. The phase transition arises because even though the particle filter has forgetting properties (Karjalainen et al., 2023), the resulting derivative estimate of Poyiadjis et al. (2011) does not, and we require both for the variance reduction.

DMOP-α achieves a lower MSE than DMOP-1, as α, k can be as large as desired to balance the impact of the last three terms of Equation 3. We also show that DMOP-0 achieves a MSE of $\tilde{O}\big(NpG'(\theta)^2(J^{-1} + (1-\epsilon)^{\lfloor 1/c\log(J)\rfloor})\big)$, where the second term corresponds to uncontrolled mixing error. Unlike DMOP-α, DMOP-0 has no opportunity to tune α (or k) to reduce said mixing error, leading to uncontrolled asymptotic bias.

8 Computational Efficiency

Our numerical results use an implementation of our methods in the `pypomp` package (Abkemeier et al., 2025). The source code for this paper is at https://github.com/pypomp/dmop. DMOP-α and IFAD are fast algorithms, both in theory and practice: the cheap gradient principle of Kakade and Lee (2018) suggests that a gradient estimate from DMOP-α should take no more than 6 times that of the runtime of the particle filter, and we found the empirical value of this factor to be 3.75. DMOP-α and IFAD therefore share the same $O(NJ)$ time complexity as the particle filter, unlike the $O(NJ^2)$ complexity of Corenflos et al. (2021) and Algorithm 2 in (Poyiadjis et al., 2011; Ścibior and Wood, 2021).

REWRITE WITH LATEST PYPOMP RESULTS Implementation of the particle filter, simulator, and DMOP-α in `JAX` (Bradbury et al., 2018) takes advantage of just-in-time compilation and GPU acceleration, even with a simulator written in Python. This led to a 16x speedup (379ms vs 6.29s on a Intel i9-13900K CPU and NVIDIA RTX3090 GPU) over the CPU-only implementation of the particle filter (with a simulator written in C++) in the `pomp` package of King et al. (2016). Additional speedup is accrued when computing multiple Monte Carlo replications, since `JAX` can take advantage of the simple parallelization opportunity.

9 Discussion

Practical likelihood-based data analysis involves many likelihood optimizations (King et al., 2008; Blake et al., 2014; Pons-Salort and Grassly, 2018; Subramanian et al., 2021; Fox et al., 2022; Drake et al., 2023). This occurs with profile likelihood confidence intervals and model selection. A careful scientist should consider many model variations, to see whether the conclusions of the study are sensitive to alternative model specifications. Coding model variations is relatively simple when using plug-and-play inference methodology. However, assessing the scientific potential of these variations requires likelihood optimization. Monte Carlo adjusted profile methods can help to acommodate maximization and evaluation error (Ionides et al., 2017; Ning et al., 2021) but these methods are most effective when the maximized log-likelihood can be obtained within a few units. Improving the process of model fitting has practical consequences for the ability of scientists to evaluate hypotheses. We have demonstrated, for the first time, a plug-and-play maximum likelihood approach which provides a high-quality log-likelihood maximization in the complex benchmark model of King et al. (2008). We anticipate that this will promote scientific advances in all domains where complex POMP models arise.

If the simulator is discontinuous in θ, DMOP-α no longer applies. We are working on a variation of DMOP-α applicable to this case. Specifically, differentiable transition density ratios can be used instead of a differentiable simulator. The off-parameter treatment of the measurement model is extended to an off-parameter simulation for the dynamic process. In that case, one requires access to the transition densities, or at least their ratios. This algorithm may be better suited to large discrete state spaces than either DMOP-α or differentiation of the Baum-Welch algorithm.

Discounting the weights by some $\alpha \in [0, 1]$ is not the only way to interpolate between the estimators of Naesseth et al. (2018) and Poyiadjis et al. (2011). As the proof of Theorem 5 implies, we can also truncate the weights at a fixed lag, corresponding to the MOP-$(1, k)$ algorithm mentioned. The analysis is similar, with comparable rates but with a slightly less convenient implementation.

Acknowledgments

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